



NSeqArpKeys

PATTERN INSTRUMENT

User manual for the pattern-driven keyboard instrument.



Assign a sequence and a Forte pitch-class set to any MIDI key. Hold the key to play the pattern, or latch it while you edit. Shape note lengths with Gate and Fixed Steps, then save the complete setup as a preset.

1. Get started

Load NSeqArpKeys as an instrument plug-in in a compatible DAW.

Install

Plug-in

Copy the complete `NSeqArpKeys.vst3` bundle to your VST3 folder and rescan in your DAW. The macOS archive also includes `NSeqArpKeys.component` for Audio Units.

Keep plug-in bundles intact. VST3 folders: Windows `C:\Program Files\Common Files\VST3` , Linux `~/.vst3` , macOS `~/Library/Audio/Plug-Ins/VST3` . For Audio Units on macOS, use `~/Library/Audio/Plug-Ins/Components` .

Make your first pattern

- 1 Open **Browse Presets**, choose **Single Note Pulse**, and click **Load**. This assigns a playable example to C4 (MIDI 60).
- 2 Hold C4 on the on-screen keyboard or send MIDI note 60 to the plug-in. The pattern plays while that key is held.
- 3 Release C4 to stop. To keep it running while editing, turn on **Latch (keep playing)**, trigger C4 once, and release it.
- 4 Click the same key again, or use **Stop Key** or **Stop All**, to stop a latched pattern.

A melodic key needs both a nonzero Pattern step and a Forte Set. Rhythmic mode uses Drum notes instead. The Init preset is deliberately silent.

2. Work in the editor

The keyboard selects a trigger note. The controls below it edit the assignment for that key. Each of the 128 MIDI keys can have its own pattern, set, channel, octave, gate, subdivision and expression.

Control	What it does
Preset name / < / >	Shows the loaded preset and steps through the preset library. An asterisk means the current setup has changed.
Presets / Pattern Bank / Save	Opens a whole-preset browser or searchable pattern bank. Save updates the current user preset, or opens the browser for a new copy.
Keyboard / Playing	Triggers a pattern and selects the MIDI key whose assignment is shown. The label below lists running keys. Right-click the keyboard to assign the selected bank pattern.
Latch / Stop Key / Stop All	Keeps patterns running after release, stops the selected key, or stops every active key.
Preview sound	Uncheck to silence the built-in tones while generated MIDI continues.
Meter Num / Global Steps/QN	Global grouping and default subdivision. Steps/QN controls the duration of one step at the current tempo.
Channel / Octave	Sets the selected key's MIDI output channel and pitch region.
Gate / Fixed Steps	Adds a gated fraction of the span through rests and a fixed duration in steps.
Pattern (length) / step preview	Space-separated step values. The label shows the number of items; the preview shows the first 16 steps. Zero creates a rest and extends the previous sounding step's gate span.
Name / Mode / Steps/QN	Name one pattern; choose melodic or rhythmic interpretation and its step subdivision. Zero inherits the global rate.
Velocity / Transpose / Rotate / Reverse	Shape output strength and pitch or change the integer step order.
Drum notes / Velocity bits/lane	In rhythmic mode enter 1-16 MIDI pitches. One numeric setting chooses the same bit width (1-7) for every pitch.
Colour / Tags / Favourite	Organize patterns visually and make them easy to find in the bank.
Cut / Clear / Copy / Paste	Remove or copy an assignment. Paste all settings, just sequence, timing, or expression.
Duplicate / Make independent	Duplicate to the next key as a separate copy, or break a shared pattern link.
Save pattern / Load pattern	Store or recall a single assignment as a <code>.nseqpattern</code> file.
Import bank / Export bank	Merge a whole pattern bank from a <code>.nseqbank</code> XML file, or save the whole bank to one file.
MIDI range / Transpose by key / Assign range	Copy an assignment into the inclusive range, optionally transposed according to each key's distance from the source.
Find Set / Forte Set	Search by Forte ID, common name, or pitch classes, then choose a matching pitch-class set.

Integer fields accept up to 4096 space-separated values, with a 45,056-character input limit. Names, tags and preset details also have length limits. Invalid input is outlined in red and is not applied. Blank Pattern clears the sequence.

Change a running pattern

With Latch on, trigger and release a key, then edit its controls while listening. Most edits restart that key on the next audio block, preserving trigger velocity. Changes to tempo or Steps/QN preserve musical phase and pending note-offs.

Selected key matters. Stop Key applies to the key shown in the selected-key label. Stop All ends every currently playing pattern.

3. Patterns, pitch, and timing

Pattern values

Enter whole numbers separated by spaces. Each number is one step. Positive numbers are bit masks over ascending notes of the selected Forte set. Negative numbers select in the opposite direction from the Octave base. Zero starts no new note.

Example - four steps

1 2 3 0

1 plays the first available note, 2 the next, and 3 plays both together. 0 creates a rest after the chord.

Octave moves the base position used by these bit masks. Channel sets the outgoing MIDI channel from 1 to 16. Notes outside MIDI's 0-127 range are ignored.

Gate across zero steps

A sounding step owns the zero steps that follow it, including zero steps at the end of a looping pattern. Gate is a percentage of that entire span, from the note onset to the next sounding step.

Example - a three-step span

1 0 0 2

The first note has three steps available: its own step and two following zeros. At 50% Gate it sounds for 1.5 steps; at 100% it reaches the next sounding step. A zero step does not retrigger the note.

Gate ranges from 0% to 200%. Values above 100% can overlap later notes. Fixed Steps ranges from 0 to 16 in increments of 0.01. The two lengths add: $\text{note length} = \text{Fixed Steps} + \text{Gate} \times \text{sounding span}$. For example, a three-step span with Gate 50% and Fixed Steps 0.5 lasts 2 steps. Releasing the trigger key stops the pattern immediately when Latch is off.

Tempo and meter

Pattern steps follow the DAW tempo when a host supplies one. Otherwise the instrument uses 120 BPM. One step lasts $60 / (\text{BPM} \times \text{effective Steps/QN})$ seconds: at 120 BPM and Steps/QN 4, a step lasts 0.125 seconds. A per-key value of 0 inherits the global setting. Meter Num is saved but does not currently change step duration or pattern length.

Rhythmic mode and variation

Rhythmic mode needs no Forte set. Enter 1-16 Drum notes; this sets the active lane count. With one velocity bit for each lane, bit 0 plays the first pitch, bit 1 the second, and so on. Try 1 0 0 0 2 0 0 0 for notes 36 and 38 with the default map. Set Channel to match the receiving instrument; changing mode preserves it.

Velocity bits/lane is one number from 1-7, shared by every active drum lane as in GateRunner. Lane 1 occupies the least significant bits of a positive decimal step mask; each later lane uses the next group of bits. A lane value of zero rests, while nonzero values scale from quiet to velocity 127. For example, two lanes with two bits each decode step 9 as level 1/3 on lane 1 and 2/3 on lane 2. Positive masks support all 16 lanes at seven bits each (up to 112 bits). Legacy signed 32-bit masks remain accepted. Base velocity and incoming key velocity further scale the result. Rotate and Reverse reorder the integer steps.

4. Forte sets

A Forte set supplies the pitch classes from which Pattern values choose notes. Each menu match shows its Forte ID, any common name, and its pitch classes in brackets.

1. Select the MIDI key you want to configure.
2. Type part of an ID, a common name such as `diatonic`, or pitch classes in **Find Set**.
3. Open **Forte Set** and choose a match. The selected set is shown underneath.

Example

7-35.11 - diatonic scale [0 2 4 5 7 9 11] contains seven pitch classes. With this set, 1 2 4 8 walks upward through its first four scale positions.

Choose **(No set)** to clear the selected key's set. A melodic pattern without a set is silent; rhythmic mode uses Drum notes independently. Changing search text only filters the menu; choose a result to change the assignment.

Live performance

Normally, a pattern runs only while its trigger note is held. Latch changes note release behavior: the first trigger keeps the pattern running, and another trigger of the same key switches it off. Turning Latch off stops patterns whose trigger keys are no longer held.

Turn off **Preview sound** to silence the built-in tones while MIDI continues. This setting is saved with the setup. Route the plug-in's MIDI output to another instrument in your DAW and match Channel to the receiving instrument. MIDI routing depends on the host.

5. Presets

A preset captures the meter, latch setting, selected key, and all 128 key assignments. The library has six factory examples and can hold your own presets.

Browse and load

Open **Browse Presets**. Search names, categories, tags, and descriptions; narrow results with the Category menu or **Favourites only**. Select a row to inspect its details, then click **Load**. Loading replaces the current setup and stops any active patterns.

Save and manage

Action	Result
Save New	Stores the current setup under the name, category, tags, and description shown on the right.
Save	Updates the currently loaded user preset. For a factory preset, opens the browser with a suggested copy name.
Update	Writes the current setup and edited details to the selected user preset. Factory presets cannot be updated.
Duplicate	Creates and loads a user copy of the selected preset.
Favourite / Unfavourite	Adds or removes the selected preset from the favourites filter.
Delete	After confirmation, removes a user preset from disk. Factory presets cannot be deleted.
Import / Export	Copies a <code>.nseqpreset</code> file into the library, or saves the selected library preset to a file. Import selects the new row; click Load to apply it.

Where presets live: in the OS application-data folder under `NSeqArpKeys/Presets` (for example, `%APPDATA%\NSeqArpKeys\Presets` on Windows). The plug-in formats share this folder on each OS.

Pattern files and whole-bank transfer

Save pattern and Load pattern use the separate `NSeqArpKeys/Patterns` folder. Pattern Bank searches names and tags, filters favourites, edits colour and details, auditions a selected pattern, and assigns it as an independent copy or shared link. Import bank and Export bank transfer all bank entries in one `.nseqbank` XML file. Import updates local entries with matching IDs, adds new entries, and retains other local entries. An entry without an ID receives one on import. Export includes all entries even when a search or favourites filter is active. Linked keys follow edits to any member; Make independent breaks the link on the selected key. Duplicate and range assignment always make independent copies. Whole presets embed shared definitions and key assignments, so they reopen without the local bank. Loading one pattern affects only the selected key. Undo/Redo lasts for the editor session and resets when loading a whole preset or project.

6. If something is not working

What you notice	What to check
No notes when a key is held	Enter a nonzero Pattern value. Melodic mode needs a Forte Set; rhythmic mode needs Drum notes and a nonzero velocity level for an active lane. Check the receiving MIDI channel.
The pattern stops when you release the key	That is the default behavior. Turn on Latch before triggering the key if you want it to continue during edits.
A pattern keeps playing	Click its trigger key again while Latch is on, use Stop Key for the selected key, or use Stop All.
VST3 does not appear in the DAW	Keep the outer <code>NSeqArpKeys.vst3</code> folder intact in the VST3 directory and rescan plug-ins in the DAW.
A preset does not appear	Check the browser's search, category, and Favourites only filters. User files must be valid <code>.nseqpreset</code> files in the preset folder.

Quick reference

To hear a pattern

Assign Pattern + Forte Set, then hold the trigger note. Use Latch to keep it running while you edit.

To share a setup

Save New, select the preset in the browser, then Export. Send the resulting `.nseqpreset` file.

NSeqArpKeys 1.3.1 | Windows, Linux, macOS | VST3; Audio Units on macOS.